Basic Information

Experiment No.: 14

Date: 11-10-2026

Title: Valid Parentheses Using Stack

Objective / Aim

To write a Java program to determine whether a string containing parentheses, curly braces, and square brackets is valid using the Stack data structure.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- IntelliJ IDEA or any Java IDE
- Operating System: Windows/Linux

Algorithm / Logic Description

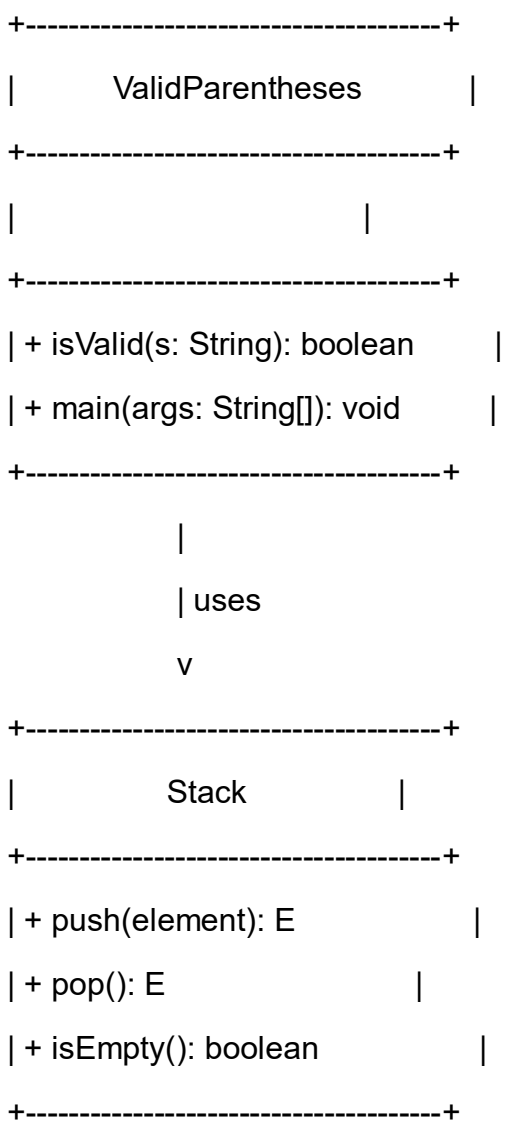
1. Read a string containing brackets.
2. Create an empty stack.
3. Push every opening bracket onto the stack.
4. For each closing bracket, check whether the stack is empty.
5. If it is empty or the bracket types do not match, mark the string invalid.

6. Otherwise, remove the matching opening bracket.

7. If the stack is empty at the end, the string is valid.

8. Display the result.

📐 UML Class Diagram



💻 Source Code

```
import java.util.Scanner;
import java.util.Stack;

public class ValidParentheses {

    // Check whether the brackets are valid
    public static boolean isValid(String s) {

        // Stack stores opening brackets
        Stack<Character> stack = new Stack<>();

        // Traverse each character in the string
        for (int i = 0; i < s.length(); i++) {

            char ch = s.charAt(i);

            // Push opening brackets onto the stack
            if (ch == '(' || ch == '{' || ch == '[') {
                stack.push(ch);
            }

            // Check closing brackets
            else if (ch == ')' || ch == '}' || ch == ']') {

                // No opening bracket to match
                if (stack.isEmpty()) {
                    return false;
```

```
            }

            char top = stack.pop();

            // Check whether the bracket types match
            if ((ch == ')' && top != '(') ||
                (ch == '}' && top != '{') ||
                (ch == ']' && top != '[')) {
                return false;
            }
        }
    }

    // Valid only when no opening brackets remain
    return stack.isEmpty();
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    // Read the input string
    System.out.print("Enter brackets: ");
    String input = sc.nextLine();

    // Display whether the brackets are valid
    if (isValid(input)) {
        System.out.println("Output: valid");
    } else {
```

```
            System.out.println("Output: Not valid");
        }

        sc.close();
    }
}
```

🔨 Compilation & Execution Commands

Save the file as:

ValidParentheses.java

Compile:

javac ValidParentheses.java

Run:

java ValidParentheses

📄 Output Screen

Example 1

Enter brackets: ()

Output: valid

Example 2

Enter brackets: ({)}

Output: Not valid

Example 3

Enter brackets:

Output: valid

📌 Result

The Java program was successfully implemented to check valid parentheses using a stack. It verifies that brackets are correctly matched and closed in the proper order, including the case of an empty string.